

Classroom AV Essentials

A Practical 2-Day Professional Development Course

“Spend less time troubleshooting. Spend more time teaching.”

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Participant Name:

School / Campus:

Date: **Location:**

Facilitator:

How to Use This Handbook

This handbook is a **permanent classroom reference** — not just a workbook you fill in and shelve. When the two-day course ends, bring it back to your classroom and keep it at your AV station, within arm's reach of your control panel.

The goal of this course is **confidence and self-sufficiency** — not turning you into an AV technician. You are a teacher first. Every skill in this handbook is chosen because it saves instructional time, reduces stress, and keeps your students learning even when technology misbehaves.

- **Use the Quick Reference Cards** (back of handbook) during a live classroom AV failure. They are designed to be torn out and laminated.

- **Use the module sections** when you want to understand *why* something works — or doesn't.
- **Use the My AV Information Page** to record your specific room's details on Day 1 or Day 2.
- **Use the Troubleshooting tables** (Modules 8–11) as your first stop when something goes wrong.

If you can consistently answer three questions — *Where is my volume control? Where is my source selector? Where is my display power button?* — you are already ahead of 80% of AV emergencies. Everything else in this handbook builds from there.

Course Schedule

Day 1 — AV Fundamentals (7 Hours Total)

Session / Module	Duration	Focus
Welcome & Introduction	30 min	Course overview, participant introductions, handbook orientation
Module 1: The AV Ecosystem	60 min	Four layers of classroom AV; key terminology; classroom walk-through
<i>Break</i>	<i>15 min</i>	—
Module 2: Audio Fundamentals	75 min	Audio chain; ANSI standards; microphone types; common audio problems
Module 3: Video and Displays	60 min	Display types; sizing formulas; display terminology
<i>Lunch</i>	<i>60 min</i>	—
Module 4: Signal Flow & Connectivity	60 min	Signal chain concept; cables and connectors; the Educator's AV Tech Kit
<i>Break</i>	<i>15 min</i>	—
Module 5: Networking Basics (Intro)	60 min	What teachers must know about AV networking; control system overview
Module 6: Accessibility & Equity	45 min	WCAG/ADA requirements; accessible AV practices; caption policy
Day 1 Wrap-Up & Kit Review	30 min	Kit assembly check; preview of Day 2; Q&A

Day 2 — Classroom AV Troubleshooting (7 Hours Total)

Session / Module	Duration	Focus
Toolkit Review	30 min	Day 1 recap; kit check; 5-Minute Rule introduction
Module 7: The Troubleshooting Mindset	60 min	Systematic vs. random troubleshooting; the 5-Step Process; when to stop
<i>Break</i>	<i>15 min</i>	—
Module 8: No Image on Screen	75 min	Room PC and laptop scenarios; quick diagnosis table; hands-on practice
Module 9: No Audio	60 min	Audio diagnosis table; restoring laptop audio; workarounds
<i>Lunch</i>	<i>60 min</i>	—
Module 10: Video Conferencing Failures	60 min	Before-call checklist; during-call problems; hybrid equity check
<i>Break</i>	<i>15 min</i>	—
Module 11: Control System Failures	45 min	Touchpanel issues; projector-specific problems; escalation criteria
Module 12: Escalation & IT Communication	45 min	Decision framework; effective ticket writing; workaround strategies
Capstone Scenario Stations	45 min	Four live troubleshooting stations; handbook-guided practice

Day 1 — AV Fundamentals

Modules 1–6 | Understanding the system before things go wrong

Module 1 — The AV Ecosystem

Duration: 60 minutes | Day 1, Morning

Every classroom AV system was designed to help you teach. When it works, it disappears into the background — you press a button and your content appears. When it fails, it costs instructional time that cannot be recovered. This module gives you the **map** of your classroom AV system so that any failure has a logical starting point for resolution.

The Four Layers of Your Classroom AV System

Layer	What It Includes	Who Owns It	Your Control Level
Source Devices	Laptop, room PC, document camera, Blu-ray/media player, streaming device	Teacher / IT	High — you use these daily
Display System	Projector, flat panel display, motorized screen, HDMI distribution	Facilities / IT	Medium — power, input, brightness controls
Audio System	Ceiling/wall speakers, microphones, amplifier, DSP/mixer	IT / AV team	Low–Medium — volume, mute, mic selection
Control System	Wall touchpanel, room controller, remote control, wall buttons	IT / AV team	Medium — UI controls; backend is IT's domain

Key AV Terms Glossary

Term	Plain English Definition
HDMI	The standard cable that carries both picture and sound together — the same type you use to connect a game console to a TV.

Term	Plain English Definition
Display / Output	Any screen that shows your content — projector, flat panel, or monitor; the destination for your signal.
Input / Source	The starting point of your content — your laptop, room PC, or document camera; what you're showing from.
Signal	The electronic information traveling from your source to your display through a cable or wireless connection.
AV Switcher	A device that lets you choose which source (laptop, PC, doc cam) appears on the display — like a channel selector for your room.
Control Panel	The touchscreen or button panel (often on the wall or podium) that operates the entire room AV system from one place.
DSP	Digital Signal Processor — the "brain" of your audio system that controls volume, echo cancellation, and microphone routing.
Codec	Hardware or software that compresses and decompresses video for conferencing — the engine behind your video calls.
Resolution	The number of pixels in the image (e.g., 1920×1080); higher resolution means more detail, but requires more processing power.
Firmware	The built-in software inside AV devices — like an operating system for your projector; IT updates it to fix bugs.
EDID	Extended Display Identification Data — the way a display tells a source what resolutions and formats it can show; EDID mismatches cause no-image problems.
HDCP	High-bandwidth Digital Content Protection — copy protection built into HDMI that can accidentally block your own content if devices are incompatible.

★ Quick Win

Right now — or during your next visit to your classroom — identify these three things:

1. Where is the **source selector** (the button or menu that chooses Room PC vs. Laptop)?
2. Where is the **volume control**?
3. Where is the **display power button**?

Knowing these three locations prevents approximately 80% of classroom AV emergencies before they become instructional disruptions.

Module 2 — Audio Fundamentals

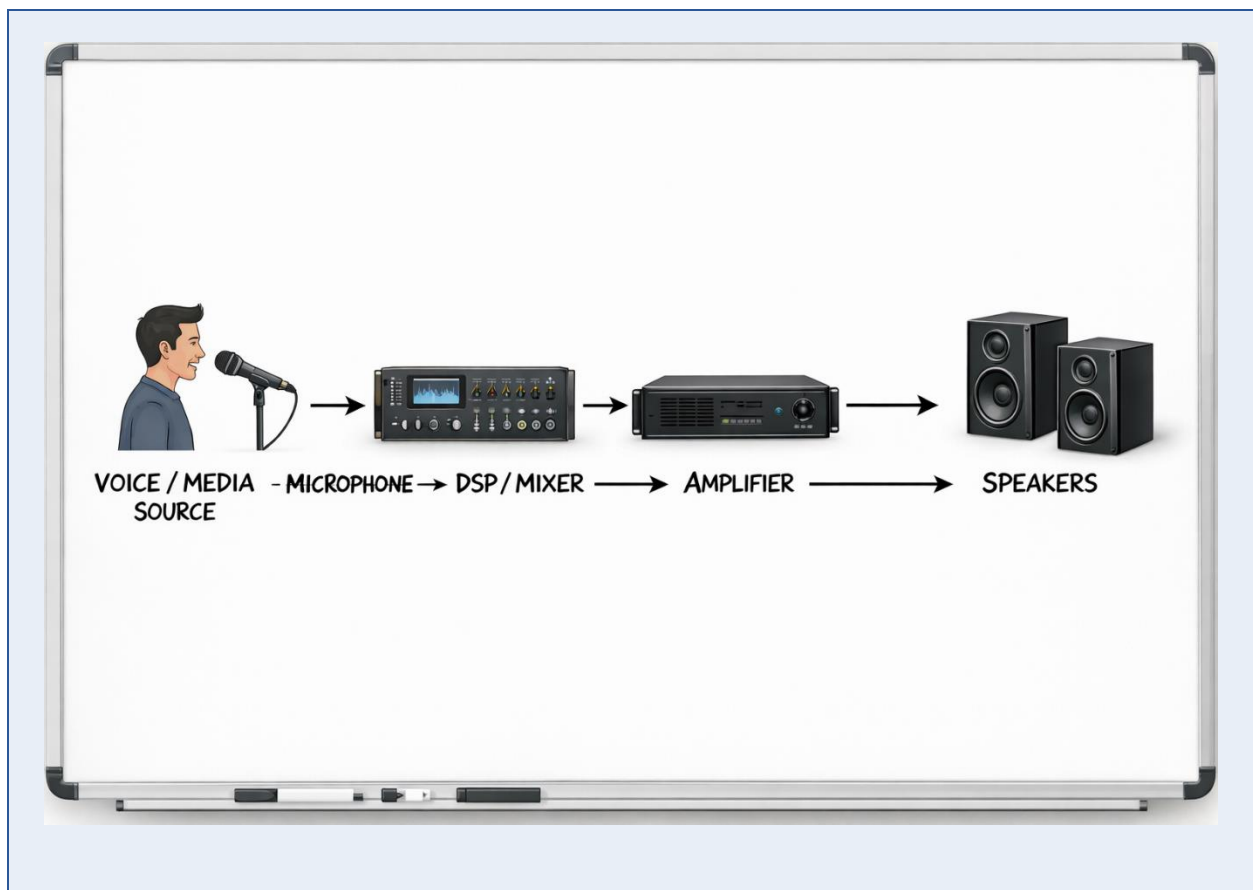
Duration: 75 minutes | Day 1, Morning

Why Audio Matters More Than Video

Poor audio — not poor video quality — is the **primary driver of student disengagement** in classroom environments. Research consistently supports what every teacher already knows intuitively: a student can follow a lesson projected on a slightly blurry or imperfectly sized screen. They **cannot** follow a lesson they cannot clearly hear. Audio is the first priority in classroom AV; when in doubt, fix sound before anything else.

The Classroom Audio Chain

Audio travels through a sequence of devices. A failure at any link breaks the chain. Effective troubleshooting means identifying *which link* broke.



1. **Voice / Media Source:** You speaking, a video playing, or music from a computer.
2. **Microphone:** Captures and converts acoustic sound to an electrical signal. Can fail if battery dead, muted, or wrong input selected.
3. **DSP / Mixer:** Processes, routes, and controls all audio signals. Settings here affect echo cancellation, EQ, and routing. This is IT's domain.
4. **Amplifier:** Boosts the processed signal to a level loud enough to drive speakers. Rarely user-serviceable.
5. **Speakers:** Convert the amplified electrical signal back into sound. A failed speaker usually sounds distorted or produces sound from only one side.

ANSI Classroom Audio Standards

Standard	Required Level	Plain English
Max Background Noise (ANSI S12.60)	≤35 dB(A)	Your room should be as quiet as a library. HVAC noise above 35 dB(A) is a reportable problem — not something to teach through.
Recommended Speech Level at Student Ear	65–70 dB	Every student, including those in the back row, should receive speech at conversational volume — which requires a properly functioning amplified sound system in larger rooms.
Max Reverberation Time (RT60)	≤0.6 seconds	Sound should "die" quickly so words don't blur into each other. Bare concrete walls and uncarpeted floors worsen reverberation — report these as design issues.

Microphone Types

Type	Description	Best Use	Common Location
Lavalier / Lapel	Small clip-on mic worn on clothing near the chest; wired or wireless (with a body pack transmitter)	Lecture-style teaching where instructor moves around the room	Instructor's lapel or collar; transmitter on belt
Boundary Mic	Flat disc mic placed on a flat surface (table, podium) that picks up sound across a wide area	Conference room tables, seminar rooms; multiple speakers around one table	Center of podium or conference table

Type	Description	Best Use	Common Location
Ceiling Array	Multi-element mic array installed in ceiling, uses beamforming to track and focus on the active speaker	Larger classrooms and conference rooms where floor space is limited	Ceiling, usually center of room
Handheld Dynamic	Traditional handheld microphone; robust, does not require phantom power; low sensitivity to handling noise	Student Q&A, presentations, multipurpose/auditorium use	Mic stand, podium, or handheld
Gooseneck	Flexible-stem mic mounted to a podium or desk; allows precise aim toward mouth	Fixed lecture podium, formal presentation settings	Podium or lectern surface

Keep Teaching

No microphone available in a class of 25 or fewer? **Project your voice and move to the center front of the room.** Position matters — a teacher standing at the center of the front wall achieves significantly better natural coverage than one standing behind a podium at the side. Your physical presence is always your most reliable audio tool and requires no IT ticket.

When to Call IT

Chronic hum, buzz, echo, or feedback every session is a system problem — not user error. If you experience persistent audio issues that return after every reboot, document the following and submit a ticket: (1) which sessions it occurs in, (2) what the sound is like (hum, buzz, echo, feedback), (3) which inputs are active, and (4) whether it happens with microphone only, media only, or both. Do not continue teaching in poor audio conditions — this is an accessibility issue.

Module 3 — Video and Displays

Duration: 60 minutes | Day 1, Before Lunch

Display Types

Type	Pros	Cons	Best Room Size
LED Flat Panel	Excellent brightness in any ambient light; sharp image; no lamp to replace; immediate on/off; long lifespan	Expensive at large sizes; fixed position; can cause glare in some room configurations	Small to medium rooms (up to ~30 ft depth); conference rooms
Standard (Lamp) Projector	Lower cost; large image possible; familiar technology; easy to repair/replace	Lamp has limited life (2,000–5,000 hrs); poor in bright rooms; warm-up/cool-down required	Medium to large rooms; rooms with reliable window shading
Laser Projector	No lamp replacement; superior brightness; 20,000+ hr light source; better in ambient light than lamp projectors	Higher upfront cost; may still require room darkening for optimal image	Medium to large rooms; auditoriums; all-day use environments
Interactive Flat Panel (IFP)	Touch interactivity; built-in annotation software; excellent image quality; all-in-one functionality	Highest cost; requires software training; touch zone limits who can interact without standing	Small to medium classrooms where instructor regularly uses interactive features

Three Numbers Every Teacher Needs

÷ 8 Minimum Display Height <i>Room Depth ÷ 8 = minimum display height</i> A 32 ft deep room needs at least a 4 ft tall display. If yours is smaller, students in the back cannot comfortably read content — this is an ADA concern.	= D ÷ W Throw Ratio <i>Projector Distance ÷ Image Width</i> A 1.5:1 throw ratio projector placed 12 ft from the screen produces an 8 ft wide image. This matters when repositioning a projector or selecting a replacement.	× 6 Max Viewing Distance <i>Display Height × 6 = max viewing distance</i> A 4 ft tall display serves students up to 24 ft away. Anyone seated beyond this distance is at a disadvantage — report the seating layout to IT or facilities.
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Display Terms Mini-Glossary

Term	Plain English Definition
1080p (Full HD)	1,920 × 1,080 pixels — the standard classroom resolution; use this as your laptop output setting for reliable compatibility with all room displays.

Term	Plain English Definition
4K / UHD	3,840 × 2,160 pixels — four times the pixels of 1080p; only useful in a 4K-capable room at close viewing distances; most classroom systems are 1080p.
Aspect Ratio 16:9	The standard widescreen shape of modern displays and projector screens — 16 units wide for every 9 units tall; matches standard PowerPoint and most content.
Lumens (ANSI)	A standardized measure of projector brightness; at least 3,500 ANSI lumens is recommended for rooms with normal window light; auditoriums may need 8,000+.
HDCP	Built-in copy protection on HDMI connections; can cause a black screen when streaming services detect it, even on content you own — power cycling all devices usually resolves this.

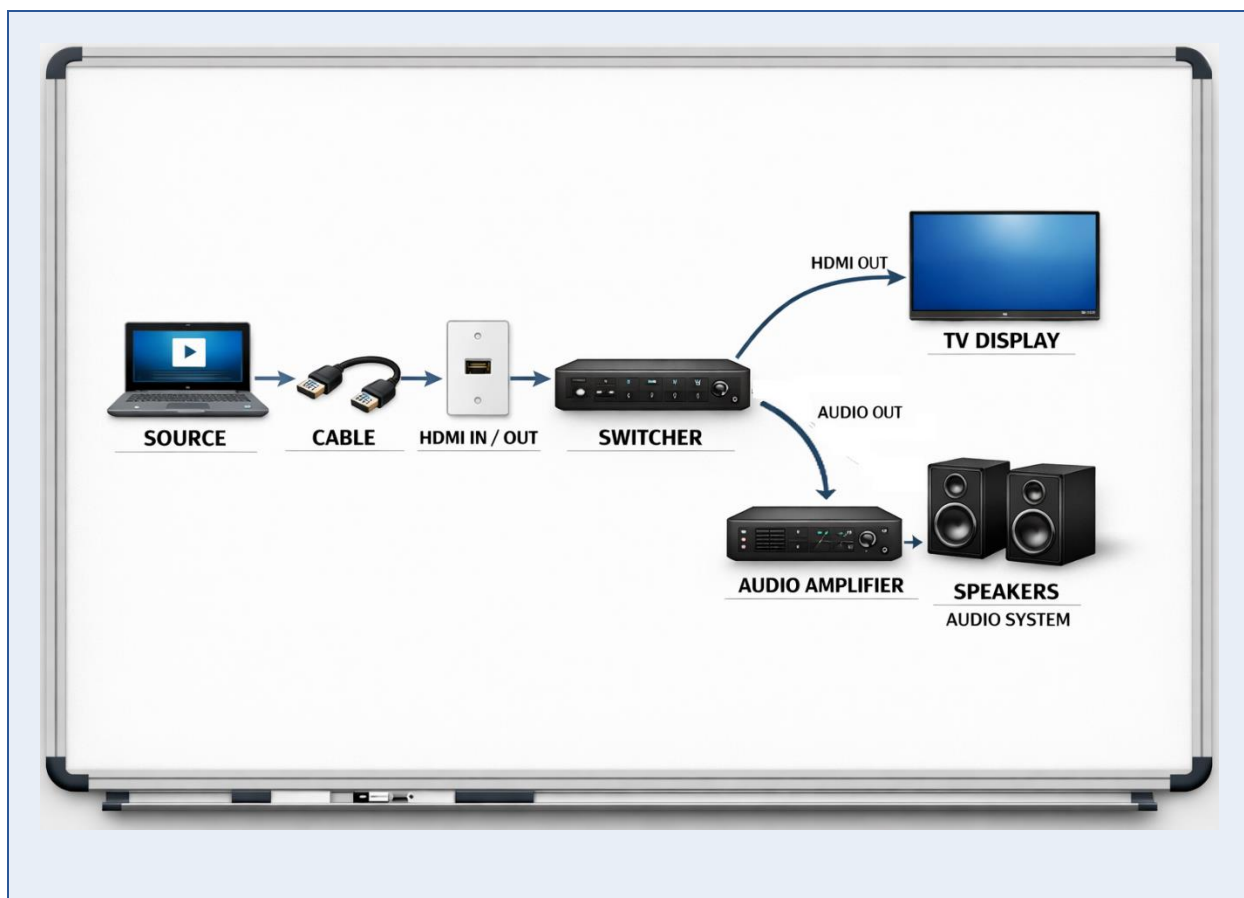
Module 4 — Signal Flow and Connectivity

Duration: 60 minutes | Day 1, After Lunch

The Signal Flow Chain

Every classroom AV problem lives somewhere on this chain. The first diagnostic question is always:

WHERE did the signal break?



Work systematically from left to right. If you can verify signal exists at a given point, move one step to the right. The break is between the last good point and the first bad point.

Cable and Connector Reference

Connector	Carries	Max Practical Length	Where Found
HDMI	Audio + Video (digital)	~15 ft passive; longer with active/fiber cable	Laptop to wall plate; room PC to display; most common classroom connector
USB-C / Thunderbolt	Audio + Video + Data + Power (digital)	~6 ft for full-spec passive; 30+ ft with active	Modern laptops (MacBooks, newer Windows laptops) to wall plate or adapter
HDBaseT / Cat6	HDMI signal over network cable (audio + video)	328 ft (100m)	Inside walls — from AV rack to wall plate; enables long runs without signal loss
VGA	Video only (analog — no audio)	~50 ft	Legacy rooms, older projectors; requires separate 3.5mm cable for audio

Connector	Carries	Max Practical Length	Where Found
3.5mm Audio	Analog stereo audio only	~25 ft	Headphone jack on laptops; often paired with VGA in older rooms
XLR	Professional balanced analog audio	200+ ft without degradation	Microphones to DSP; professional audio equipment; 3-pin round connector
USB-A / USB-B	Data, device control (not AV signal)	~16 ft passive	Document cameras, webcams, room controllers, laptop to AV system USB connections

The Educator's Proactive AV Tech Kit — Full Inventory Checklist

Build this kit once. Keep it at your AV station. It costs approximately **\$40–\$60 total** and prevents hours of lost instructional time per semester.

Section 1 — Essential Cables and Power

- ☐ **HDMI cable, 10 ft** — your primary video/audio cable backup
- ☐ **USB-C to USB-A cable** — for connecting modern laptops to older USB ports
- ☐ **IEC power cord (projector/display backup)** — the standard computer-style power cable; projectors and displays often share this type

Section 2 — Adapters and Dongles

- ☐ **USB-C to HDMI adapter** — essential if your laptop uses USB-C output
- ☐ **HDMI to VGA adapter** — for use in legacy rooms with VGA-only wall plates only
- ☐ **USB-C to USB-A adapter** — for connecting peripherals to USB-C-only laptops

i Important Note

Keep **only the adapters that match your actual equipment**. Do not mix digital and analog signal paths — for example, do not use a USB-C → HDMI → VGA chain; signal quality degrades and HDCP conflicts become more likely.

Section 3 — Power / Batteries

- ☐ **AA batteries (4+)** — for remotes and wireless mic body packs
- ☐ **AAA batteries (4+)** — for smaller remotes and some wireless mic receivers

i Battery Rule

Check remote batteries **first** when a control system appears unresponsive. Dead batteries cause approximately 30% of apparent control system failures and take 10 seconds to diagnose.

Section 4 — Quick Reference Guides

- ☐ **This handbook**
- ☐ **Laminated Quick Reference Card** (produced in Day 2 — tear out from back of handbook)
- ☐ **IT Help Desk number** posted visibly at the AV station

★ Quick Win

Label your kit with your name and room number. AV kits migrate. A labeled kit returns to its owner. Total kit cost: approximately \$40–\$60. Lost instructional time prevented per semester if the kit saves you from even one 30-minute AV failure: \$40–\$60 is an exceptional return on investment.

Module 5 — Networking Basics for Teachers

Duration: 60 minutes | Day 1, Afternoon

What You Don't Need to Know

You do not need to understand TCP/IP addressing, VLAN configuration, QoS settings, subnet masks, or network topology. **That is IT's job.** What you do need is enough knowledge to communicate accurately with IT, recognize when a network is causing an AV problem, and avoid actions that worsen the situation.

Five Things Every Teacher Should Know About AV Networking

6. **AV devices have IP addresses.** Projectors, displays, DSPs, and control systems all connect to your network with an IP address. If a device is rebooted and receives a different IP address than expected, the control system will show it as "offline" even though the physical device is powered on. When this happens, tell IT the specific device name and the symptom — don't try to reassign addresses yourself.
7. **HD video calls need 3–8 Mbps upload AND download symmetrically.** If your video conferencing call is choppy, pixelated, or drops, the first diagnostic question is whether other network-heavy activities are happening simultaneously on the same circuit. Ask students to close streaming apps during video calls.
8. **AV devices and student devices are often on separate networks intentionally.** This is a security measure. If you cannot "see" a wireless display or AV device from your laptop, this is the most likely reason — it is not a failure. Work with IT to ensure your device is on the correct network segment for AV access.
9. **Firmware updates fix many chronic problems.** If you have a recurring, unexplained AV issue that IT cannot reproduce, ask specifically whether all devices in your room are running current firmware. This is one of the most frequently overlooked fixes for intermittent failures.
10. **Wired Ethernet is always preferable to Wi-Fi for AV.** For video conferencing, content sharing, and control system communication, a physical Ethernet cable eliminates the single largest source of AV network variability. Treat Wi-Fi as a backup, not a primary AV connection strategy.

Control System Reference

What You See	What It Does	Your Control Level
Touchpanel (wall or podium)	Master control interface for all room AV — power, source selection, volume, mic routing, display presets	Full — this is your primary tool
Room PC / Laptop via HDMI	Content source; OS-level display and audio settings controlled here	Full — Win+P, display settings, audio output selection
AV Switcher / Matrix	Routes multiple sources to multiple displays; usually controlled via touchpanel	Medium — via control panel UI only; hardware is IT-managed
DSP / Amplifier	Processes all audio signals, applies echo cancellation, controls routing	Limited — volume and mute via control panel only; all DSP settings are IT's domain
Firmware and IP Settings	Backend software and network configuration for all AV devices	IT Only — never attempt to access device admin interfaces without IT authorization

Module 6 — Accessibility and Equity

Duration: 45 minutes | Day 1, Afternoon

Every AV decision you make affects which students can access your instruction. **Accessible AV is not a special accommodation — it is good teaching.** Captions help English language learners, students with auditory processing differences, students in noisy environments, and every student who reviews recordings later. Proper display sizing helps students with low vision and also helps every student in the back row. Accessible AV benefits every learner, every session.

Minimum Accessibility Practices Checklist

- ☐ **Enable closed captions on every video by default.** Make "always on" your standard, not an exception.
- ☐ **Use the room microphone whenever available.** If an amplified sound system is present, use it — even in a small class. Students with hearing differences often cannot self-identify.

- ☐ **Meet minimum contrast on displayed text.** WCAG 2.1 AA requires a 4.5:1 contrast ratio for body text. White text on a bright yellow background fails. White on dark blue passes.
- ☐ **Avoid red/green distinctions as the only means of conveying information.** Approximately 8% of males have color vision deficiency; red/green is the most common form. Add labels, patterns, or shapes as secondary indicators.
- ☐ **Minimum 24pt body text / 36pt key points for projected content.** Test every slide from the back row of your room before using it in class.
- ☐ **Never show an uncaptioned video unless no captioned version exists** — and if none exists, provide a transcript or written summary.
- ☐ **Know where assistive listening system receivers are stored.** Ask IT or facilities today if you don't know.
- ☐ **Report accessibility failures immediately.** These are civil rights issues under ADA Section 504. A non-functioning assistive listening system is not a normal IT ticket — it is an accommodation failure with legal implications.

ADA Quick Reference

Requirement	Standard	Teacher Action
Background Noise Level	≤35 dB ANSI S12.60	Close doors during instruction. Report loud HVAC, noisy equipment, or external construction noise to facilities — these are reportable conditions, not facts of life.
Assistive Listening Systems (ALS)	Required in all amplified assembly spaces (ADA Section 4.33)	Know where ALS receivers are stored and how to issue them. Ensure the system is functional at the start of any session where an accommodation requires it.
Captions	FCC and ADA require captions for broadcast television; best practice extends to all video content in educational settings	Enable captions as a default setting on all media playback. Never require a student to disclose a hearing difference to request captions — just keep them on.

Requirement	Standard	Teacher Action
Control Panel Height	Maximum 48 inches AFF (Above Finished Floor) per ADA	If your control panel is inaccessible from a seated position (e.g., mounted at 60+ inches), report it to facilities as an ADA barrier — do not work around it as normal.

Keep Teaching — Captions Are for Everyone

Captions benefit: students who are Deaf or hard of hearing; English language learners; students with auditory processing differences; students in temporarily noisy rooms; every student who reviews recorded content later. **Make captions the default. Always on.** The only cost is two seconds of configuration.

Day 1 Wrap-Up — Preview of Day 2

Tomorrow is entirely about what happens when things go wrong — and what you do while waiting for IT. You will leave Day 2 with:

- A laminated **Quick Reference troubleshooting card** for your AV station
- A **completed kit checklist** specific to your room
- A **completed My AV Information page**
- Confidence to handle approximately **80% of classroom AV failures independently**
- A clear protocol for the 20% that requires IT escalation

Day 2 — Classroom AV Troubleshooting

Modules 7–12 | What to do when things go wrong — and how to keep teaching while they're wrong

Module 7 — The Troubleshooting Mindset

Random button-pressing wastes time, can worsen some problems, and gives IT inaccurate information when you eventually call. **Systematic troubleshooting is a learnable, repeatable skill** — the same mental process used by AV technicians, IT professionals, and engineers. You do not need to become an expert. You need the process.

Before You Touch Anything — Answer Four Questions

11. **What EXACTLY is wrong?** — "No image" is different from "wrong image." "No sound" is different from "distorted sound." Be specific.
12. **What is connected?** — Which source? Which cable? Which input selected on the control panel?
13. **When did it start?** — Beginning of class? After a specific action? Intermittently for weeks?
14. **What changed recently?** — New cable? Different laptop? OS update? Someone else used the room?

The 5-Step Classroom Troubleshooting Process

Step	Action	What to Do
1	STOP AND OBSERVE	What are you seeing — or not seeing? What are status LEDs doing (green = on/good; amber = standby/warning; red = error; blinking = communicating or fault)? Write down what you observe before touching anything.
2	FOLLOW THE SIGNAL	Source → Cable → Switcher → Display. Ask: Is the source on and outputting? Is the cable fully connected at both ends? Is the switcher showing the right input? Is the display powered and on the right input? Work left to right.
3	TRY SIMPLE THINGS FIRST	Power cycle the display (not the room PC). Re-seat the HDMI cable. Select the correct input on the control panel. Press Windows+P → Duplicate. Check remote batteries. In that order.
4	TEST ONE VARIABLE AT A TIME	Swap one cable OR one adapter OR one input — never all at once. If you change three things simultaneously, you cannot know which change fixed the problem. This makes future failures harder to diagnose and impossible to document accurately for IT.
5	KNOW WHEN TO STOP	If the problem is not resolved in 5 minutes — STOP troubleshooting. Switch to a workaround AND submit an IT ticket simultaneously. Instructional time is not recoverable. The lesson continues; the AV problem gets resolved later.

THE 5-MINUTE RULE

Give yourself exactly 5 minutes on any AV problem.

If not fixed in 5 minutes — **STOP.**

Switch to a workaround. **TEACH.**

Submit an IT ticket at the next break.

Instructional time is not recoverable. Every minute spent troubleshooting past 5 minutes is a minute taken from your students' learning.

Escalate Immediately — Do NOT Troubleshoot

⚠ Stop and Escalate — No Exceptions

- ☐ Burning smell or any visible damage to equipment or cables
- ☐ Sparking, flickering power, or any sign of electrical fault
- ☐ Open rack cabinet with accessible live equipment
- ☐ Any person harmed by electrical equipment
- ☐ Control system completely unresponsive after full reboot (2+ minutes wait)

For any safety item: clear students from the area, call facilities and/or security first, then IT.

Module 8 — No Image on Screen

Duration: 75 minutes | Day 2, Morning

Scenario A — Room PC: No Image on Screen

15. **CHECK DISPLAY POWER:** Look at the power LED on the display or projector. Solid green or blue = powered on and active. Amber, blinking, or off = the display is not in active mode. Press the power button on the control panel (not on the display itself). Wait 15 seconds for the display to wake fully.
16. **SELECT CORRECT INPUT:** On the control panel, select the input labeled "COMPUTER" or "HDMI 1." If you see "LAPTOP," that is a different input. Do not guess — the label matters.
17. **TOGGLE DISPLAY OUTPUT (WIN+P):** On the room PC keyboard, press Windows key + P. A sidebar appears. Select "Duplicate." Wait 5 full seconds for the display to re-detect the signal.
18. **CHECK HDMI CABLE AT PC:** Locate the HDMI cable at the back of the PC tower. Unplug it, wait 3 seconds, then plug it back in firmly — you should feel a slight click. Visually inspect for bent pins before reinserting.
19. **REBOOT PC (last resort):** Reboot the room PC with the HDMI cable already connected. A cold boot forces the operating system to re-detect the display on startup. This takes 2–4 minutes — set students to an independent activity before initiating.

Keep Teaching — Room PC Failure

Connect your personal laptop to the "LAPTOP" input while IT addresses the room PC. Have your class materials available on both devices or accessible via the LMS/cloud. Room PC failure is IT's problem; your lesson does not have to be.

Scenario B — Laptop: No Image on Screen

20. **CHECK DISPLAY POWER:** Same as Scenario A — confirm the display is on and not in standby.
21. **SELECT CORRECT INPUT:** On the control panel, select "LAPTOP" or "HDMI 2." Laptop and room PC are on different inputs — selecting the wrong one is the single most common cause of a no-image scenario.

- 22. CONNECT THE CABLE:** Fully seat the HDMI or USB-C cable at both ends — at the laptop and at the wall plate. A half-seated connector is the most common physical cause of no image.
- 23. TOGGLE DISPLAY OUTPUT:** Windows: Win+P → Duplicate. Mac: Apple menu → System Settings → Displays → check "Mirror Displays." Wait 5 seconds after each selection.
- 24. ADAPTER CHECK:** If you are using a USB-C or other adapter, try a different adapter if one is available in your kit. Adapters fail silently — they provide no error message; the connection simply does not work.
- 25. RESOLUTION FIX:** If an image appears but is the wrong size, has black bars, or is zoomed incorrectly — right-click the desktop → Display Settings → set resolution to 1920×1080 → Apply. This resolves the majority of EDID mismatch symptoms.

No-Image Quick Diagnosis Table

Symptom	Most Likely Cause	Quick Fix
Black screen (display power light is on)	Wrong input selected on control panel	Select correct input: Room PC → "COMPUTER/HDMI 1"; Laptop → "LAPTOP/HDMI 2"
"No Signal" text on screen	Cable not connected or not seated	Re-seat HDMI cable at both ends; confirm correct input selected
HDCP error message or unexplained black screen	Copy protection conflict between source and display	Power cycle all devices in sequence: source off → display off → wait 10 sec → display on → source on
Correct image appears but wrong size / resolution	EDID mismatch between laptop and display	Right-click desktop → Display Settings → set to 1920×1080
Image visible on laptop but not on room display (extended desktop)	Display mode set to "Extend" instead of "Duplicate"	Win+P → Duplicate (Windows) or Mirror (Mac)
Image appears but is very dark or washed out	Projector lamp is dim or display brightness at minimum	Raise display brightness via control panel; if projector, submit lamp warning ticket
Signal drops intermittently (image flickers on and off)	Failing or damaged HDMI cable	Swap cable with the 10 ft HDMI from your kit; submit ticket if problem resolves (confirms failing cable)

Module 9 — No Audio

Duration: 60 minutes | Day 2, Before Lunch

Audio Problem Quick Diagnosis Table

Symptom	Most Likely Cause	Quick Fix
No sound at all from room speakers	Control panel is muted or volume at zero	Press Mute button on control panel to toggle off mute; raise volume slider
No sound from laptop (image is fine)	Laptop audio is routed to its internal speakers, not to room system	Click taskbar speaker icon → click the chevron/arrow → select HDMI Output or display name (see steps below)
No sound from a specific video or browser	In-application mute or volume at zero	Check browser tab speaker icon, YouTube player volume, and media player volume — all three can be muted independently
Microphone not working	Wrong microphone input selected on DSP or control panel	Select correct mic input on control panel; check wireless mic battery and power switch
Feedback / squeal from speakers	Microphone too close to speakers or gain too high	Move mic away from speakers; lower mic volume on control panel; if chronic, submit ticket
Low, constant hum or buzz	Ground loop between laptop charger and room audio system	Disconnect laptop power charger temporarily and test; if hum stops, report to IT — this is a system design issue
Echo on video calls (remote participants hear themselves)	Acoustic Echo Cancellation (AEC) not functioning	Use a headset immediately for the remainder of the call; submit IT ticket — do not attempt to adjust DSP
Audio from one side of the room only	Speaker failure or amplifier channel failure	Note which side (left, right, front, back); submit ticket; use portable speaker as workaround

Restoring Laptop Audio to Room Speakers — Windows

26. Click the **speaker icon** in the Windows taskbar (lower right corner).

27. Click the **chevron or arrow** (▸) next to the volume slider — this expands the output device list.

28. Select the output labeled "**HDMI Output**," "**Display Audio**," or your room display name from the list.

29. **Test with a short audio clip** — open a browser and play 5 seconds of any video. If you hear audio in the room, the fix worked.

Restoring Laptop Audio to Room Speakers — Mac

30. Click the **Apple menu** () → **System Settings**.

31. Click **Sound** in the sidebar → select the **Output** tab.

32. Select the output labeled "**HDMI**" or your display name.

33. Adjust volume using the volume slider on this screen or the keyboard.

Keep Teaching — No Room Audio

Option 1: A Bluetooth portable speaker charged overnight works for most classroom audio content including video. Keep one in your AV kit. **Option 2:** Your voice is always your most reliable audio source. If showing a video, narrate the key points while it plays silently — this is also a pedagogically strong active-engagement technique. Submit the IT ticket at the next break.

Module 10 — Video Conferencing and Hybrid Failures

Duration: 60 minutes | Day 2, After Lunch

Before-the-Call Checklist

Complete this checklist **5 minutes before every video conference begins**. Never assume it will work because it worked last week.

- ☐ **Join 5 minutes early — always.** Problems discovered at meeting start time cannot be fixed at meeting start time.

- ☐ **Test audio preview.** Use the platform's audio test feature — speak and verify your voice is captured; play a test tone and verify you hear room audio.
- ☐ **Test video preview.** Confirm the correct camera is selected and the image is clear and well-lit.
- ☐ **Test content sharing.** Open the document or slide you will share and do a test share before participants join.
- ☐ **Have your laptop fallback ready.** Personal laptop open, logged into the platform, headset plugged in. If room AV fails, you can continue the call within 60 seconds.

During-Call Problem Diagnosis

Problem	Most Likely Cause	Quick Fix
Call drops immediately upon joining	Bandwidth limit or firewall blocking the platform	Switch to wired Ethernet immediately; ask IT if the platform is allowed on the network
Remote participants hear echo of their own voice	Room microphone is picking up room speaker output	Use a headset for the remainder of the session; report AEC failure to IT
Cannot hear remote participants	Wrong audio output selected in platform settings	Open platform audio settings → output device → select HDMI or room speakers by name
Camera shows black screen	Wrong camera selected in platform settings, or camera covered	Platform settings → video/camera → select correct camera by name; remove privacy cover if present
Content share shows black screen to participants	HDCP conflict on a streaming source via HDMI	Share from a different source (e.g., open the file directly in a browser); avoid sharing streaming platforms via HDMI-connected sources
Captions not appearing	Captions not enabled in platform	Find the CC (Closed Captions) or Live Transcript button in the platform toolbar and activate it; enable by default in platform settings
Video freezes for participants (audio continues)	Insufficient upload bandwidth	Lower video quality in platform settings; switch to audio-only mode; close all other network-intensive applications

Hybrid Equity Quick Check

Before beginning any hybrid session, verify equity for remote participants by answering these five questions:

- 34. Can remote participants **hear every in-room speaker** — including students asking questions?
- 35. Can remote participants **see the relevant area of the room** (board, display, instructor, student presenter)?
- 36. Can remote participants **read all shared on-screen content** clearly and without lag?
- 37. Can remote participants **participate equally** — hand raise, chat, polling, Q&A — without being talked over?
- 38. Is **captioning on by default** — not waiting to be requested?

If any answer is "no," resolve it before beginning instruction. Hybrid equity is not optional — it is the definition of accessible learning.

Module 11 — Control System and Other Common Failures

Duration: 45 minutes | Day 2, Afternoon

Control System Quick Reference

Symptom	Most Likely Cause	Quick Fix	Escalate?
Touchpanel screen is black / off	Sleep mode (normal behavior)	Touch the screen to wake; if no response after 10 sec, press and hold the home/power button for 3 sec	Yes — if no response after 30 sec
Touchpanel is on but buttons do nothing	Control processor is rebooting (takes 60–90 sec)	Wait 60–90 seconds without pressing anything — repeated pressing can interrupt the boot sequence	Yes — if unresponsive after 2 full minutes
Input selected on panel but display did not switch	Firmware fault or signal interruption in switcher	Tap the same input again; then power cycle the display only (not the control panel)	Yes — if problem repeats in same session

Symptom	Most Likely Cause	Quick Fix	Escalate?
"System Unavailable" or similar message	Control processor has gone offline	Note the exact message; wait 60 sec and retry; do not press repeatedly	Yes — call IT help desk directly
One specific function fails; everything else works	An individual AV device has lost its IP address or connection	Note exactly which function failed (e.g., "microphone mute button does nothing"); submit a specific ticket	Yes — submit ticket with device name

Projector-Specific Issues

Symptom	Most Likely Cause	Quick Fix
Lamp warning light (usually orange/red) is on	Lamp is approaching end of life or overheating	Submit a ticket immediately — do not ignore lamp warnings. Continue using until IT advises; note that response time may need to be same-day for critical rooms.
Yellow or green color tint across entire image	Lamp is degrading (color temperature shift)	Submit ticket; continue using if image is still readable. A color-shifted image is an accessibility issue — escalate accordingly.
Fan running unusually loudly	Filter is clogged or projector is overheating	Submit ticket; check that nothing is blocking the projector's air intake or exhaust vents
Projector shuts off unexpectedly mid-class	Thermal shutdown (overheating protection activated)	Clear all objects from around the projector immediately; allow 15 minutes to cool before restarting; submit ticket; use display workaround during cool-down
Black bars on left/right or top/bottom edges of image	Overscan setting or resolution mismatch	Set laptop resolution to 1920×1080; if problem persists, submit ticket requesting overscan/EDID adjustment

Module 12 — Escalation and IT Communication

Duration: 45 minutes | Day 2, Afternoon

The Escalation Decision Framework

Question	If YES	If NO
Is this a safety issue? (smoke, sparks, heat, electrical fault, person harmed)	Stop immediately. Clear students from area. Call facilities / security first. Then IT. Do NOT troubleshoot.	Continue to next question.
Resolved in 5 minutes or fewer?	Fix and document for IT (brief note). Continue teaching.	Activate workaround now. Submit IT ticket at next break. Continue teaching.
Does it affect an accessibility accommodation?	Call IT help desk directly AND document the accommodation impact in writing. Do not use standard ticket-only workflow.	Standard ticket is appropriate. Include all details per checklist below.

What to Include in an AV IT Ticket — Checklist

- ☐ **Room number and building**
- ☐ **Date and time** the problem occurred
- ☐ **Exact symptom** — describe what you see/hear (e.g., "No image on projector when laptop connected via HDMI through USB-C adapter")
- ☐ **What you tried** — list every step taken in order
- ☐ **What is connected** — exact cable type, adapter brand/model if known, which wall plate input was used
- ☐ **Intermittent or consistent** — does it happen every time, or randomly?
- ☐ **Accessibility accommodation impact** — if any accommodation was affected, state this explicitly
- ☐ **Your contact info and availability** for IT to reach you or visit the room

Ticket Quality — Side-by-Side Comparison

✗ Poor Ticket — Do Not Write This	✓ Effective Ticket — Write This
"Projector broken again. Please fix."	"Room 204, Building A. Wednesday 9:05 AM. Projector shows 'No Signal' when MacBook Pro (M2) connected via USB-C to HDMI adapter (from my kit) to HDMI 2 wall plate input. Tried: re-seating cable, Win+P Duplicate, different adapter. Problem is consistent every session for 2 weeks. No accommodation impact. Available Mon–Wed 3–4 PM or email. — P. Brown, 204A"

Workaround Strategies — Keep Teaching Table

AV Failure	Keep Teaching Workaround	Time to Implement
No projector/display image	Share slides via LMS on student devices; post PDF link in chat or LMS announcement	~2 minutes
No room audio	Portable Bluetooth speaker; narrate video content while it plays silently	~1 minute
Video conferencing failure	Phone call for audio; share slides via LMS; continue via personal laptop hotspot if approved	~3 minutes
Room PC failure	Personal laptop connected to LAPTOP input on control panel	~2 minutes
Control system unresponsive	Use manual controls directly on the display (power and input buttons on the display itself)	~1 minute
No microphone	Project voice and move to center-front of room; reduce class size impact by moving students closer	Immediate
Document camera offline	Phone camera on a stand connected via USB or Bluetooth screen mirroring; photograph and share via LMS	~3 minutes
Total AV failure (all systems down)	Whiteboard + printed handouts; pivot to discussion-based or collaborative activity; this is a facilities emergency — call immediately	Immediate

Capstone Activity — Scenario Stations

Duration: 45 minutes | **Format:** Rotate through 4 stations, approximately 10 minutes each

At each station you will find a scenario card describing a real classroom AV failure. Using your handbook and your kit, work through the 5-Step Troubleshooting Process to diagnose, resolve (or escalate) the problem. Record your findings in the spaces below.

Station 1 — "The Blank Screen"

Scenario: Room PC is on. HDMI 2 selected on the control panel. Display power light is solid green. No image appears on screen — the screen is completely black.

Symptom (your words):	
First thing checked:	
Root cause identified:	
Fix OR workaround used:	
Escalate? (Y/N + why):	
IT ticket wording:	

Station 2 — "The Silent Video"

Scenario: Laptop is connected via HDMI. Image is displaying correctly on room screen. You play a YouTube video — the video plays and you can see it moving, but there is no sound from the room speakers.

Symptom (your words):	
First thing checked:	
Root cause identified:	
Fix OR workaround used:	
Escalate? (Y/N + why):	
IT ticket wording:	

Station 3 — "The Frozen Call"

Scenario: You are in a video conference call. The call freezes every 20 seconds — the video pixelates and stops briefly. Remote participants report they can hear you clearly but your video keeps dropping out.

Symptom (your words):	
First thing checked:	
Root cause identified:	
Fix OR workaround used:	
Escalate? (Y/N + why):	
IT ticket wording:	

Station 4 — "The Unresponsive Panel"

Scenario: The touchpanel on the wall is lit up and the screen is on. When you press any button — volume, input, anything — nothing happens. You need to switch to the laptop input immediately.

Symptom (your words):	
First thing checked:	
Root cause identified:	
Fix OR workaround used:	
Escalate? (Y/N + why):	
IT ticket wording:	

Quick Reference Cards

Tear out along the dotted border. Laminate. Keep at your AV station.

CARD 1 — NO IMAGE ON SCREEN

Use this card when: display is powered on but shows black screen, "No Signal," or wrong image.

☐ **STEP 1 — DISPLAY POWER:** Power LED green/blue = ON. Amber or blinking = press Power on control panel. Wait 15 sec.

☐ **STEP 2 — CORRECT INPUT:** Room PC → select "COMPUTER" or "HDMI 1." Laptop → select "LAPTOP" or "HDMI 2."

☐ **STEP 3 — WIN + P → DUPLICATE:** Windows: Win+P → Duplicate. Mac: System Settings → Displays → Mirror. Wait 5 sec.

☐ **STEP 4 — RE-SEAT HDMI CABLE:** Unplug at both ends. Wait 3 sec. Reconnect firmly. Check for bent pins.

☐ **STEP 5 — TRY A DIFFERENT CABLE:** Use the 10 ft HDMI cable from your kit.

☐ **STEP 6 — REBOOT PC (last resort):** Reboot with HDMI already connected. Wait 2–4 min. Do NOT reboot display.

STILL BROKEN? Switch to workaround. Submit IT ticket. Keep teaching.

IT Help Desk: Room:

CARD 2 — NO AUDIO

Use this card when: room speakers are silent, microphone not working, or unexpected echo/feedback.

- ☐ **STEP 1 — CONTROL PANEL MUTE:** Is the Mute button on? Press it to toggle off. Raise volume slider to 50–70%.
- ☐ **STEP 2 — LAPTOP AUDIO OUTPUT:** Click taskbar speaker icon → click arrow (>) → select HDMI Output or display name.
- ☐ **STEP 3 — IN-APP VOLUME:** Check browser tab mute, YouTube player volume slider, and media player volume — all three independently.
- ☐ **STEP 4 — MIC INPUT:** Confirm correct microphone is selected on control panel. Check wireless mic battery and power switch.
- ☐ **STEP 5 — ECHO:** If echo — use headset immediately. Submit ticket. Do not adjust DSP settings.
- ☐ **STEP 6 — HUM / BUZZ:** Unplug laptop charger temporarily and test. If hum stops → submit ticket (ground loop issue).

STILL BROKEN? Use portable speaker. Narrate video. Keep teaching.

IT Help Desk: Room:

Glossary of AV Terms

Alphabetical reference. Plain English definitions for classroom use.

Term	Plain English Definition
AEC	Acoustic Echo Cancellation — software processing that prevents a microphone from picking up speaker output and looping it back to remote participants; when AEC fails, remote callers hear their own echo.
AV Switcher	A device that selects which source (laptop, PC, document camera) is sent to the display and audio system; usually controlled via the room's touchpanel.
Codec	Hardware or software that compresses video for transmission and decompresses it at the receiving end; the engine inside your video conferencing system.
DSP	Digital Signal Processor — the audio processing unit that controls volume, equalization, echo cancellation, microphone routing, and mixing for the room's sound system.
EDID	Extended Display Identification Data — the digital "handshake" between a source device and a display that establishes what resolutions and formats are supported; EDID mismatches cause no-image or wrong-resolution symptoms.
Firmware	Embedded software inside AV devices (projectors, DSPs, control processors) that controls their operation; firmware updates fix bugs and add features — ask IT if a device has a chronic problem.
Ground Loop	An electrical phenomenon that occurs when two pieces of equipment are connected to power at different ground potentials; causes a constant low hum or buzz in audio output; resolved by IT.
HDCP	High-bandwidth Digital Content Protection — copy protection embedded in HDMI connections that can accidentally block legitimate educational content; power cycling all devices in sequence resolves most HDCP conflicts.
HDMI	High-Definition Multimedia Interface — the standard cable and connector for carrying digital audio and video together; the most common AV connection in classrooms.
HDBaseT	A technology that transmits HDMI signals over standard network cable (Cat5e/Cat6) for distances up to 328 feet; used inside walls to connect AV racks to wall plates in classrooms.
Hybrid Equity	The principle that remote participants in a hybrid class receive an equivalent learning experience to in-room participants — including equal audio, visual, and participation access.
QoS	Quality of Service — a network configuration that prioritizes video conference traffic over other data; managed by IT; if video calls are consistently poor, ask IT to verify QoS settings.

Term	Plain English Definition
Resolution	The pixel count of a video signal (e.g., 1920×1080); higher resolution means more detail; set your laptop to 1920×1080 (1080p) for reliable compatibility with all classroom displays.
RT60	Reverberation Time — the time it takes for a sound to decay by 60 dB in a room; ANSI S12.60 requires RT60 ≤0.6 seconds in classrooms; longer RT60 means speech sounds muddy and unclear.
Signal Flow	The path an AV signal travels from source to display/speakers; understanding signal flow is the foundation of all AV troubleshooting — find the break in the chain.
Throw Ratio	The ratio of projector distance from the screen to the width of the image produced (distance ÷ image width); used to determine projector placement and image size.
VLAN	Virtual Local Area Network — a method of dividing a physical network into separate logical networks; AV devices are often on a separate VLAN from student devices for security and performance reasons.
VGA	Video Graphics Array — an older analog video connector (usually blue, 15-pin) that carries video only (no audio); found in legacy AV rooms and older projectors; requires a separate audio cable.
XLR	A professional 3-pin audio connector used for microphones and balanced audio equipment; the circular connector at the end of a professional microphone cable; resistant to interference over long cable runs.

My AV Information Page

Complete this page on Day 1 or Day 2. Ask your IT contact or AV support team for device-specific details.

Keep this page current — update it after any room equipment change.

School / Campus	
Building and Room Number	
IT Help Desk Phone Number	
AV / Tech Support Phone Number	
Control System Type / Brand	
Display Type (Projector / Flat Panel / IFP)	
Display Size (diagonal inches)	
Input Label for Room PC	
Input Label for Laptop	
Input Label for Document Camera (if present)	
Microphone Type(s) in Room	
Wireless Mic Battery Type	
Assistive Listening System Type	
Assistive Listening Receivers Located At	
Known Room Quirks (anything IT has told you)	
Firmware Last Updated (ask IT for date)	
Last IT Visit / Service Date	

Acknowledgments

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Content is developed in alignment with **WCAG 2.1 Level AA** accessibility guidelines, **ANSI S12.60** classroom acoustics standards, **ADA Sections 504 and 508** requirements, and **FCC captioning regulations** as applicable to educational environments.

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